

Blockchain and Application Stack

L5 Presentation

React 18 + Vite,
MetaMask, ethers v6



L4 API & Services

Express / Node 20 ·
FastAPI · WebSocket
EIP-712 auth · JWT · rate-
limit



L3 Data

PostgreSQL 16 (3NF, 15+
tables) · Redis 7 · IPFS
doc store



L2 Smart-Contract Platform

Solidity 0.8.20 ·
OpenZeppelin v5 · UUPS ·
TimelockController ·
RBAC · ReentrancyGuard



L1 Settlement / Network

Polygon PoS (retail) ·
Ethereum Sepolia
(institutional)
Chainlink CCIP bridge ·
The Graph indexer ·
Tenderly monitor